TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
Linear combinations: make one variable's coefficients opposites, then add the equations.
Multiply every term, including the number on the right side.
Fractions: multiply by the LCD first. Variable alone? Substitution is faster.

1. Use linear combinations to solve $x + y = 9$ and $x - y = 3$.

Tip: Add the equations: y and −y drop out.

A. (3, 6)

B. (12, −3)

C. (6, 3)

2. You add $2x + 5y = 4$ and $-2x + 3y = 12$. Which variable drops out?

Tip: Look for opposite coefficients.

F. y

G. both

H. x

3. Solve: $2x + 5y = 4$ and $-2x + 3y = 12$.

> **Tip:** Add the equations first.

Add: ____y = 16, so y = _____

2x + 5(____) = 4

x = _____